Step IV: Blocking

The nitrocellulose membrane is non-specifically saturated or masked by using casein or Bovine serum albumin (BSA) before adding the primary antibody. This blocking step is very important in western blotting as antibodies are also proteins and they are likely to bind to the nitrocellulose paper.

Step V: Treatment with primary and secondary antibody

The primary antibody is specific to desired protein so it forms Ag-Ab complex. The secondary antibody is enzyme labelled and is against primary antibody (anti-antibody) so it can bind with Ag-Ab complex. Alkaline phosphatase or Horseradish peroxidase (HRP) is labelled with secondary antibody.

Step VI: Treatment with suitable substrate

Finally, the reaction mixture is incubated with specific substrate. The enzyme convert the substrate to give visible coloured product, so band of colour can be visualized in the membrane (Figure 11.6).

Application

1. The size and concentration of protein in given sample is determined by western blotting.
2. It is used in the detection of antibody against virus or bacteria in serum and helps in the disease diagnosis.
3. Western blotting technique is the confirmatory test for HIV. It detects anti HIV antibody in patient's serum.
4. Useful to detect defective proteins.

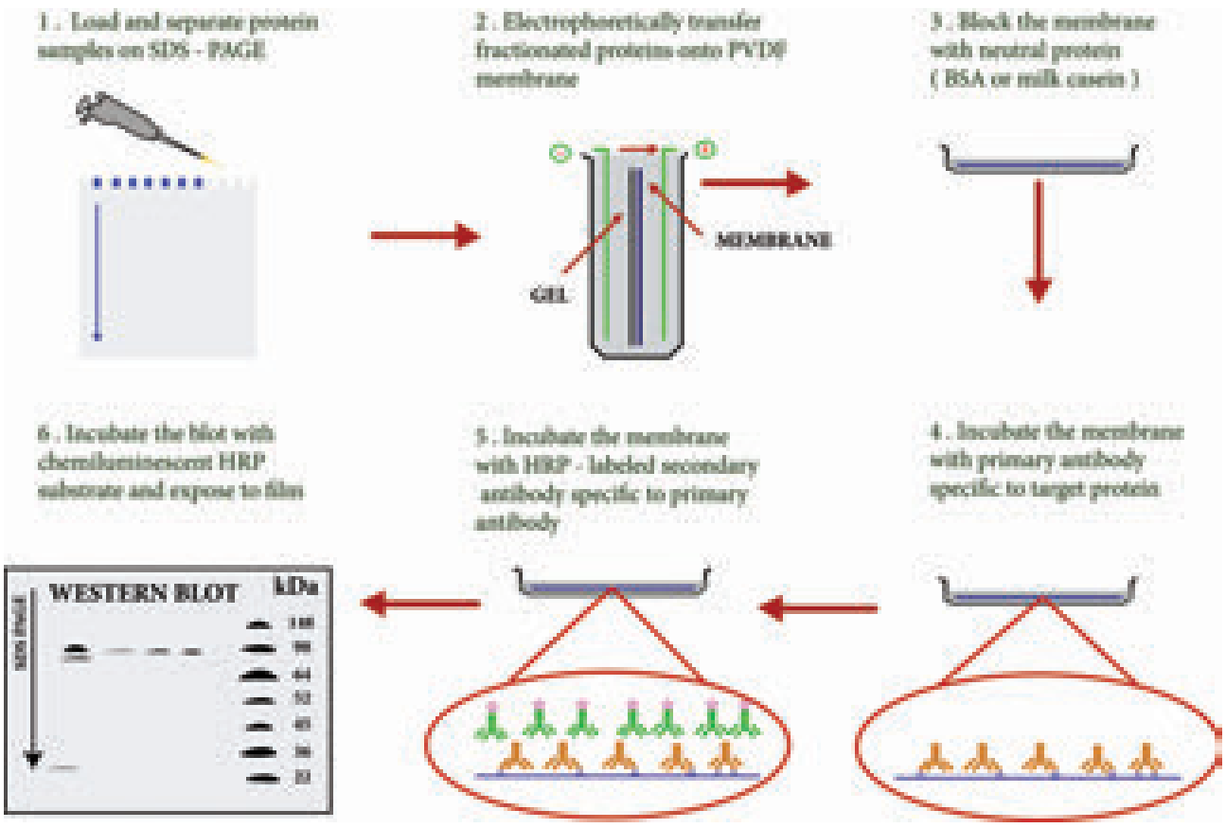


Figure 11.6: Western blot technique

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